RAJENDRA KUMAR

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SUMMARY

Healthcare Data Scientist with 6+ years building machine learning, image analysis, and statistical models on medical imaging, clinical text, and claims data. Experienced in statistical analysis, time series forecasting, and taking algorithms from feasibility and prototype to production in HIPAA-regulated environments.

EDUCATION

Master of Science in Data Science (Big Data Systems) — Aug 2023 – May 2025
Indiana University Bloomington — GPA: 3.7

PG-Diploma in Big Data Analytics & Machine Learning — Feb 2018 – Aug 2018
Centre for Development of Advanced Computing, Bangalore — Grade: A

Bachelor of Engineering in Computer Science & Engineering — Jun 2013 – Jul 2017
Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal — GPA: 8.02

WORK EXPERIENCE

AI Data Engineer – Segmed Inc. (Remote – Bloomington, IN, USA) — Nov 2025 – Present

- Prototype and build agentic AI system automating feasibility assessments, scaling from 400–500/year to 300/month.
- Develop Python/SQL pipelines to process and validate DICOM medical imaging and structured clinical records.
- Streamline imaging workflows with monitoring, fast debugging, and HIPAA-aligned PHI de-identification.
- Collaborate cross-functionally with engineering, clinical, and product teams to deliver curated research datasets.

Senior Consultant (AI/ML) – Heartland Network — Jun 2025 – Nov 2025

- Built time series forecasts and presented findings on clinical and employment trends to leadership and clients.
- Performed EDA, preprocessing, and feature engineering, merging diverse data sources into model-ready datasets.
- Developed interactive KPI dashboards tracking staffing (physicians, RNs) and daily/monthly patient visits per site.
- Mentored junior team members, guiding their analysis and modeling work on client projects.

Research Assistant – FADS – Indiana University Bloomington — Jan 2025 – May 2025

- Fine-tuned Stable Diffusion 1.5 (LoRA/QLoRA) and GANs on custom image datasets; tracked 5+ models in W&B.
- Integrated GPT-4V for image description; deployed models via Docker with a React/Flask/PostgreSQL web app.

Senior Data Scientist – Target Corporation — Apr 2023 – Jun 2023

- Extended in-house computer vision model classifying shoplifting behavior (concealing items), achieving 87% F1 score.
- Created real-time KPI dashboards (Domo, Greenfield, Power BI) guiding budget allocation and inventory buys.
- Built fraud detection model (Random Forest, class-imbalance handling): 0.97 ROC AUC, 0.99 precision, 0.80 recall.

Sutherland Global — Sep 2018 – Mar 2023

Data Scientist (Sep 2018); promoted to Senior Data Scientist (Jun 2020) and Lead Data Analytics – Data Science (Jun 2021)

- Delivered Propensity-To-Pay (P-T-P) predictions via dashboards and a real-time page used by 3,000+ specialists.
- Mentored junior data scientists on ML methods and Python/SQL programming practices.
- Forecasted claims flow with ARIMA, RNN, and LSTM models, cutting pre-filing claim review time and cost by 30%.
- Enhanced ensemble model predicting insurance claim actions and status codes, reaching 95% accuracy.
- Analyzed claim denial patterns to surface revenue cycle insights, increasing client collections by 15%.
- Prototyped Smart-Doc image processing tool (PyTorch, OpenCV) for scanned EHR/PHI documents at 94% accuracy.
- Engineered pipelines converting XLSX, CSV, PDF, and image files into HL7 format for healthcare data exchange.
- Tracked parameters, metrics, and artifacts for 20+ ML models using MLflow on Azure Databricks.

SKILLS

Certifications	**AWS ML Associate**, **AWS AI Practitioner**, **Azure AI**, **Databricks Gen AI**
Programming	Python, R, SQL, Java, Shell scripting
Imaging & Signal	OpenCV, PyTorch, TensorFlow, CNN, nnU-Net, 3D segmentation, OCR, DICOM, NIfTI, ARIMA
Statistics & ML	Hypothesis testing, A/B & causal lift tests, ROC/AUC, ensemble models, Scikit-learn, XGBoost
Data & MLOps	PySpark, Hive, PostgreSQL, MongoDB, MLflow, W&B, Docker, Git, AWS, Azure Databricks, GCP
Healthcare & Tools	HL7, EHR/PHI, ICD/CPT/SNOMED, HIPAA de-identification, Tableau, Power BI, MS Office

PROJECTS

Propensity-To-Pay (Predictive Modeling): *Tech Used: Python, SQL, PySpark, Hive, Random Forest, XGBoost*

- Devised PySpark scripts to process transaction data into account-level aggregates, storing in Cloudera Hive tables.
- Led P-T-P algorithm design from scratch to production, reaching 93% accuracy via GridSearchCV tuning.

Clinical Text – Knowledge Extraction (NLP): *Tech Used: Python, spaCy, NLTK, BERT, ClinicalBERT, UMLS*

- Built clinical NER pipeline (spaCy, NLTK, ClinicalBERT) assigning ICD/CPT/SNOMED codes at 88% accuracy.
- Extracted drug entities, relations, and smoking status from high-volume clinical text, storing results in MongoDB.

Brain Metastasis Segmentation – 3D MRI: *Tech Used: Python, PyTorch, nnU-Net, NiBabel, NumPy, Pandas*

- Trained 3D residual-encoder U-Net on 1,295 MRI cases to segment tumor subregions and resection cavity.
- Ensembled 5-fold cross-validated models and tuned connected-component postprocessing for 0.76 average Dice.
- Built NIfTI quality checks for cross-modality geometry (affine, spacing) and lesion-level error analysis.